

A CERTIFIED NORMALIZED FLOW FOR THE ABSOLUTE ORDER OF TYPE D_9

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ABSTRACT. Gaetz and Gao conjectured that the absolute order of every Coxeter group of type D_n admits a normalized flow, and reported a computer verification through $n = 8$. We settle the next finite instance. We quotient $\text{Abs}(D_9)$ by conjugation by B_9 , classify the resulting 150 vertices by signed cycle type, reconstruct all 609 supported adjacent-rank orbit pairs, and give 284 nonzero rational orbit-flow values. Uniform lifting over each biregular orbit-pair cover graph produces a normalized flow on the full poset. The primary JSON certificate and its frozen verifier use only exact integer and rational arithmetic. A separately written no-import referee verifier reconstructs the full quotient from signed permutations, checks all nine rank layers and 3,344,302,080 covers, and rejects nine targeted certificate corruptions. This is a certified result for D_9 only; it does not prove the conjecture for arbitrary n .

1. INTRODUCTION

For a finite reflection group W with reflection set T , the absolute length $\ell_T(w)$ is the least number of reflections whose product is w . The absolute order is

$$u \leq_T v \iff \ell_T(u) + \ell_T(u^{-1}v) = \ell_T(v).$$

Gaetz and Gao conjectured that $\text{Abs}(D_n)$ has a normalized flow with unit vertex weights [2, Conjecture 1.3(a), p. 792]. They reported a computer search through $n = 8$, without publishing the corresponding code or certificates [2, final paragraph of Section 5, p. 799].

The purpose of this paper is to certify the first instance beyond that published range.

Theorem 1.1. *The absolute order $\text{Abs}(D_9)$ admits a normalized flow with unit vertex weights.*

The result is finite and exact. In particular, “first” here means the first integer beyond the published $n \leq 8$ report. Two dated public literature and code searches, the second performed after the certificate was found, located no public resolution as of August 29, 2026; this bounded statement makes no assertion about private or unindexed computations.

The main difficulty is size: $|D_9| = 2^8 9! = 92,897,280$. The exact certificate avoids expanding all elements. Conjugation by B_9 is a poset automorphism

Date: August 29, 2026.

action, and its 150 orbits in D_9 are indexed by signed cycle types. Each adjacent-rank normalized-flow problem then becomes a rational transportation problem. We solve all nine layers with integer max flow, serialize the rational totals, and independently reconstruct every datum needed to lift them to the full poset.

2. THE FINITE ENDPOINT

Write $B_9 = C_2 \wr S_9$ for the signed permutation group in its natural action on $V = \mathbb{R}^9$. In window notation, let

$$D_9 = \{w \in B_9 : \text{the window of } w \text{ has an even number of negative entries}\}.$$

For $1 \leq i < j \leq 9$, the reflection set T contains the unsigned signed transposition and the signed transposition negating both exchanged entries:

$$(i \ j)(-i \ -j), \quad (i \ -j)(-i \ j).$$

Thus $|T| = 2\binom{9}{2} = 72$.

Let $P_r = \{w \in D_9 : \ell_T(w) = r\}$. A normalized flow with unit vertex weights is a family of nonnegative values $f_r(x, y)$, one for every cover $x \lessdot y$ from P_r to P_{r+1} , such that

$$(1) \quad \sum_{y: x \lessdot y} f_r(x, y) = \frac{1}{|P_r|}, \quad \sum_{x: x \lessdot y} f_r(x, y) = \frac{1}{|P_{r+1}|}.$$

This is the normalization used by Gaetz and Gao [2, Section 2.4, p. 794]. The nine adjacent-rank systems are independent, zero flow is allowed on a cover, and no rank symmetry is assumed.

3. SIGNED-CYCLE QUOTIENT

3.1. The automorphism action.

Lemma 3.1. *Conjugation by B_9 acts on $\text{Abs}(D_9)$ by rank- and cover-preserving automorphisms.*

Proof. The product of the window signs is a homomorphism $B_9 \rightarrow \{\pm 1\}$ whose kernel is D_9 , so $D_9 \triangleleft B_9$. A signed coordinate permutation sends every root $\pm e_i \pm e_j$ to another root of this form. It therefore permutes exactly the 72 reflections in T . Conjugation hence preserves ℓ_T , the defining equality for absolute order, and covers. $\square$

This action is intentionally larger than conjugation by D_9 . Consequently any D_9 -conjugacy classes that split inside a single B_9 -class are re-fused by a genuine automorphism action; no split class is lost.

3.2. Orbit types, masses, and ranks. A signed cycle is positive or negative according to the product of its signs. A B_9 -conjugacy class is described by a bipartition (λ, μ) , where λ lists its positive-cycle lengths and μ its negative-cycle lengths. The class lies in D_9 precisely when

$$(2) \quad |\lambda| + |\mu| = 9, \quad \ell(\mu) \equiv 0 \pmod{2}.$$

For a partition α , put $z_\alpha = \prod_j j^{m_j(\alpha)} m_j(\alpha)!$.

Lemma 3.2. *The B_9 -orbits in D_9 are the 150 types satisfying (2). The orbit $C_{\lambda, \mu}$ has size and rank*

$$(3) \quad c(\lambda, \mu) = \frac{2^9 9!}{2^{\ell(\lambda) + \ell(\mu)} z_\lambda z_\mu}, \quad \text{rank } C_{\lambda, \mu} = 9 - \ell(\lambda).$$

Proof. The signed-cycle decomposition gives the bipartition classification, while the total sign is $(-1)^{\ell(\mu)}$. For m signed cycles of a fixed length j and fixed sign, the centralizer contributes $(2j)^m m!$: there are $2j$ internal signed rotations and the equal cycles may be permuted. Multiplication over all parts gives centralizer order $2^{\ell(\lambda) + \ell(\mu)} z_\lambda z_\mu$ in B_9 , proving the mass formula.

Carter defines reflection length using the root reflections and proves that it is the number of eigenvalues different from 1 [1, Lemma 2, p. 3]. His reflections are exactly the 72 elements of T , because the D_9 roots are $\pm e_i \pm e_j$ and roots differing by sign give the same reflection. A positive signed cycle has a one-dimensional fixed space and a negative signed cycle has none. Disjoint cycle supports form a direct sum, so $\dim \text{Fix}(w) = \ell(\lambda)$ and Carter's formula yields $\ell_T(w) = 9 - \ell(\lambda)$.

Finally, exhaustive partition generation subject to (2) gives 150 types. All divisions in (3) are integral, and their masses sum to 92,897,280, independently checking exhaustiveness. $\square$

The exact rank orbit counts and full rank sizes are respectively

$$(4) \quad (|\mathcal{O}_0|, \dots, |\mathcal{O}_9|) = (1, 1, 3, 5, 11, 17, 29, 35, 34, 14),$$

$$(|P_0|, \dots, |P_9|) = (1, 72, 2220, 38304, 405174, 2702448,$$

$$(5) \quad 11228300, 27491616, 34812945, 16216200).$$

These ranks are not symmetric, and all nine layers are checked directly.

3.3. Covers and their multiplicities. Right multiplication by a reflection either joins two signed cycles or splits one. An upward cover decreases the number of positive cycles by one, so the complete list is

$$P_a + P_b \longrightarrow P_{a+b}, \quad P_a + N_b \longrightarrow N_{a+b}, \quad P_{a+b} \longrightarrow N_a + N_b.$$

If m_a^+ and m_b^- are the source multiplicities, the number of reflections carrying a fixed source element to the indicated target type is

$$(6) \quad d_{A,B}^+ = \begin{cases} 2ab m_a^+ m_b^+, & P_a + P_b \rightarrow P_{a+b}, \ a \neq b, \\ a^2 m_a^+ (m_a^+ - 1), & P_a + P_a \rightarrow P_{2a}, \\ 2ab m_a^+ m_b^-, & P_a + N_b \rightarrow N_{a+b}, \\ (a+b) m_{a+b}^+, & P_{a+b} \rightarrow N_a + N_b, \ a < b, \\ am_{2a}^+, & P_{2a} \rightarrow N_a + N_a. \end{cases}$$

For a merge, one selects the source cycles and one of the $2ab$ signed transpositions joining their supports. For a split of a positive $(a+b)$ -cycle, an unordered cut has $a+b$ positions when $a < b$ and a positions when $a = b$; exactly one of the two sign choices makes both new cycles negative. This proves (6) and exhausts the three rank-increasing sign possibilities.

Let $E(A, B)$ be the number of full-poset covers between two quotient nodes. Lemma 3.1 makes the induced bipartite graph biregular. If $d_{A,B}^-$ is the target-side degree, then

$$(7) \quad E(A, B) = c(A) d_{A,B}^+ = c(B) d_{A,B}^-.$$

The independent referee does not trust (6): it instead builds a canonical signed permutation of every type and multiplies it by all 72 reflections from both endpoints. This separately reconstructs support, d^+ , d^- , and (7).

4. EXACT QUOTIENT FLOW AND LIFTING

For $A \in \mathcal{O}_r$ and $B \in \mathcal{O}_{r+1}$ joined by covers, let $F(A, B)$ denote the total quotient flow between the two orbits.

Lemma 4.1. *Suppose nonnegative rational numbers $F(A, B)$ satisfy*

$$(8) \quad \sum_B F(A, B) = \frac{c(A)}{|P_r|}, \quad \sum_A F(A, B) = \frac{c(B)}{|P_{r+1}|}$$

for every adjacent rank pair and every orbit in that pair. Then assigning $F(A, B)/E(A, B)$ to each cover between A and B gives a normalized flow on $\text{Abs}(D_9)$.

Proof. For a fixed $x \in A$, the contribution from the orbit pair (A, B) is

$$d_{A,B}^+ \frac{F(A, B)}{E(A, B)} = \frac{F(A, B)}{c(A)}.$$

Summing over B and using (8) gives $1/|P_r|$. The same calculation from the target side gives $1/|P_{r+1}|$. $\square$

This is the orbit-quotient equivalence of Gaetz and Gao specialized to the present action [2, Proposition 3.6, p. 797].

To construct F exactly, let

$$L_r = \text{lcm}(|P_r|, |P_{r+1}|), \quad s_A = L_r \frac{c(A)}{|P_r|}, \quad d_B = L_r \frac{c(B)}{|P_{r+1}|}.$$

These are integers with total supply and demand L_r . A bipartite network on the supported quotient edges therefore reduces each layer to an integral max-flow problem. Dividing a saturating integer flow by L_r gives the rational values $F(A, B)$. No floating-point feasibility result is used.

5. CERTIFICATE AND INDEPENDENT VERIFICATION

The unique primary theorem certificate is `certificates/d9_normalized_flow.json`; its SHA-256 is

`d8fe947906c577f3a7c585fc07e0bcb6648e2555056a756af98bf307ab72ea7f`.

It contains the 150 orbit records and the 284 positive values $F(A, B)$; the other 325 supported orbit pairs have zero flow. The frozen primary verifier is `certificates/verify_d9_flow.py`, with SHA-256

`a076f7dc4a715d0adb9d1c7b7946431a7f63de744b53a8782a6c50ed7580c706`.

The roles and hashes are fixed in `certificates/PRIMARY_D9_BINDING.json`. A second JSON encoding and checker are retained only as a cross-check and are not interchangeable with the primary pair.

Table 1 gives the complete layer summary. The full Hasse diagram has 609 supported orbit pairs and 3,344,302,080 covers.

TABLE 1. Exact D_9 certificate summary by adjacent ranks.

ranks	supported pairs	positive F	L_r	full covers
0–1	1	1	72	72
1–2	3	3	13,320	5,112
2–3	9	7	7,086,240	154,728
3–4	22	15	369,518,688	2,603,160
4–5	50	27	26,070,515,856	26,569,368
5–6	97	45	7,585,974,219,600	168,006,888
6–7	156	65	1,641,936,765,600	640,430,712
7–8	175	72	106,340,457,307,680	1,338,965,640
8–9	96	49	12,545,192,860,200	1,167,566,400
total	609	284	—	3,344,302,080

The primary verifier reconstructs the bipartitions, orbit masses, ranks, cover support, multiplicities, edge counts, rational marginals, and the uniform F/E lift. It also clears each rational layer by L_r and checks integer conservation. An independently written no-import referee verifier, `audit/referee_verify_d9.py`, uses only the Python standard library and the primary JSON. It imports or executes no discovery program, test module, or primary verifier. Instead it rebuilds canonical signed permutations and all reflections, reproducing exactly the counts in Table 1.

The referee verifier rejects duplicate or unknown JSON keys, floats, non-canonical fractions, missing structural data, and invalid endpoints. Its

mandatory mutation suite rejects all nine targeted corruptions: a hash mismatch, a missing layer, a missing flow, a wrong-rank endpoint, an altered degree, an altered edge count, a negative rational, a non-reduced rational, and an altered rank size. Its final unresolved severity counts are `fatal=0`, `major=0`, `local=0`, and `expository=0`.

Proof of Theorem 1.1. Lemmas 3.1 and 3.2 give the exact B_9 orbit quotient and its ranks. Equations (6) and (7), independently reconstructed by the referee verifier, give every supported adjacent-rank orbit pair. The primary certificate supplies nonnegative rational $F(A, B)$ whose row and column marginals (8) are checked exactly in every layer. Applying Lemma 4.1 gives values on every cover satisfying (1). Hence $\text{Abs}(D_9)$ admits the asserted normalized flow. $\square$

6. REPRODUCIBILITY, SCOPE, AND DISCLOSURE

From the release root, the two decisive checks are

```
python3 -I -S certificates/verify_d9_flow.py \
  certificates/d9_normalized_flow.json
python3 -I -S audit/referee_verify_d9.py
```

Both use exact integers and `fractions.Fraction`. The second command also runs the nine mutation tests on every successful invocation. The serialized manifest binds the formal statement, primary certificate, verifiers, manuscript, and release artifacts by SHA-256.

The theorem concerns only D_9 . It asserts neither a recurrence from D_n to D_{n+1} nor a closed flow formula for arbitrary n , uniqueness, strict positivity on every cover, or rank symmetry. The published D_8 boundary was separately reconstructed as a baseline, but no result for $n > 9$ is used or claimed.

Computational and AI assistance was used to discover the quotient flow, write verification code, organize source searches, and prepare the manuscript. The mathematical endpoint is bound to serialized exact data and two fail-closed verifiers, one of which was written under a no-import trust boundary. No Lean, Coq, Isabelle, or other proof assistant was used; no machine-checked formalization is claimed.

REFERENCES

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